

Digital Camera Specification Analysis: Trends in Camera Technology and Manufacturer Differences (1994–2007)

Executive Summary

This project analyzes the evolution of digital camera specifications between 1994 and 2007 using the *1000 Camera Dataset* obtained from Kaggle. The objective was to examine how camera technology changed over time and identify differences among manufacturers by analyzing variables such as manufacturers, release year, megapixels, and camera weight. The data set was cleaned and explored using Google Sheets through pivot tables, scatter plots, line charts to identify trends and relationships. The analysis found that average camera resolution increased steadily throughout the study period. While average weight of cameras steadily declined, suggesting that during this period manufacturers produced increasingly compact cameras without sacrificing image quality. Manufacturer comparisons also revealed differences in average camera resolution and weight, with some companies specializing in higher resolution or heavier camera models. Due to concerns regarding the reliability of the dataset's price variable, price was excluded from the final analysis. Overall the findings illustrate how digital camera technology evolved during this period and demonstrate the use of exploratory data analysis to identify meaningful trends within historical product data.

Business Task

The objective of this project was to analyze the historical digital camera specifications between 1994 and 2007 to identify technology trends and compare manufacturers. Using exploratory data analysis, the project investigates how camera specifications evolved over time and how manufacturers differed in the products they released during this period.

To answer the objective, the following business questions were explored:

1. Which manufacturers produced cameras with the highest average effective megapixels between 1994 and 2007?
2. How did camera resolution change between 1994 and 2007?
3. Do heavier cameras tend to have higher effective megapixel counts?
4. How did average camera weight change between 1994 and 2007?
5. Which manufacturers produced the heaviest cameras on average between 1994 and 2007?

Dataset

The dataset used for this analysis was the 1000 Camera Dataset, obtained from Kaggle. It contains specifications for 1,000 digital cameras released between 1994 and 2007 by 21 different manufacturers. Variables included manufacturer, model, release year, effective megapixels, maximum resolution, camera weight (including the battery), dimension, zoom specifications and other technical characteristics.

For this project, the variables analyzed were:

- Manufacturer
- Release year
- Effective Megapixels
- Weight (including battery)

These variables were selected because they directly support answering the business questions regarding technological advancements and manufacturer differences during this study period.

Data Cleaning and Preparation

The data was imported into Google Sheets for cleaning and for exploratory analysis. Before beginning the analysis the data was reviewed to make note of missing values, inconsistencies, and variables relevant to the projects business questions.

The following preparation steps were completed:

- Verified the dataset for missing or incomplete values in the variables used for analysis. This was done through conditional formatting and various google sheets tools.
- Reviewed variable names and data types to ensure numerical variables could be summarized and visualized correctly.
- Created PivotTables to calculate average effective megapixels and average camera weight by manufacturer and release year.
- Created scatter plots to examine the relationship between camera weight and effective megapixels.
- Created line charts to evaluate changes in average camera resolution and average camera weight over time.
- Created horizontal bar charts to compare manufacturers based on average effective megapixels and average camera weight.

During the exploratory analysis, the price variable was analyzed but excluded from the final analysis. While the dataset contained a column labeled "Price," repeated values and comparisons with historical camera specifications suggested that this variable did not reliably represent original camera prices. Because the dataset documentation did not clearly define the variable, it

was removed from the analysis to avoid drawing unsupported conclusions. The analysis therefore focused on variables with clearly defined and internally consistent measurements.

Analysis

Business Question 1

Which manufacturers produced cameras with the highest average effective megapixels between 1994 and 2007?

Average Effective Megapixels by Manufacturer (1994-2007)

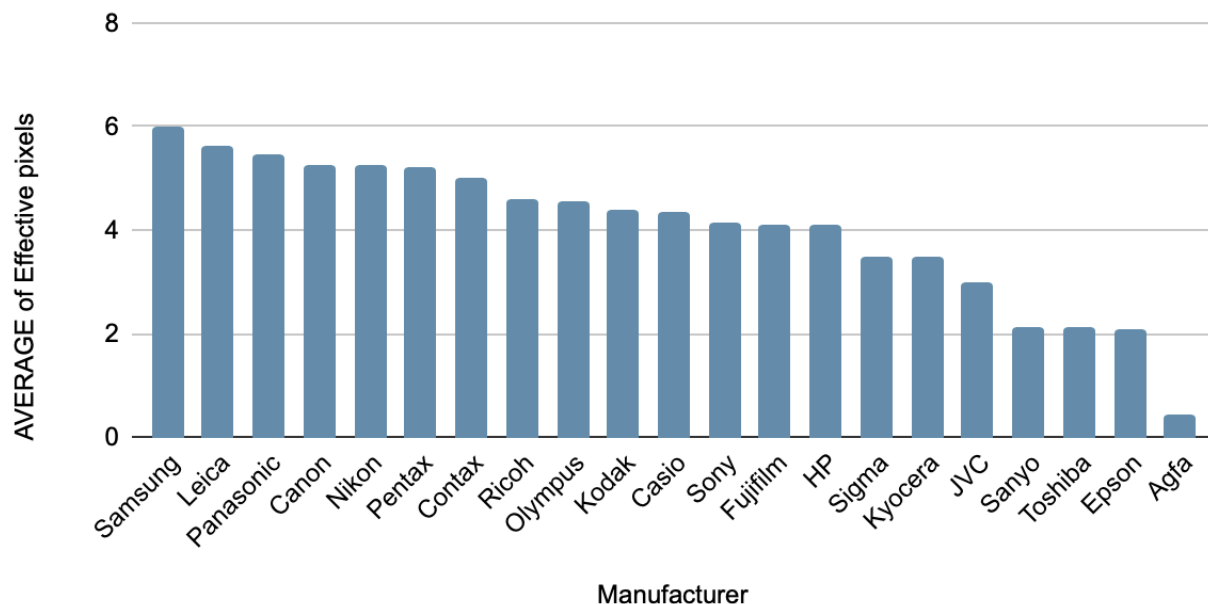


Figure 1

Findings

The average effective megapixels varied across manufacturers between 1994 and 2007, although the differences were relatively modest. Samsung had the highest average effective megapixel count, followed closely by Leica, Panasonic, Canon, Nikon, and Pentax. Agfa had the lowest average effective megapixel count among the manufacturers included in the dataset. Overall, most manufacturers averaged between approximately four and six effective megapixels, indicating that many companies offered cameras with similar average image resolutions during this period.

Business Question 2

How did camera resolution change between 1994 and 2007?

Average Effective Megapixels by Release Date

(1994-2007)

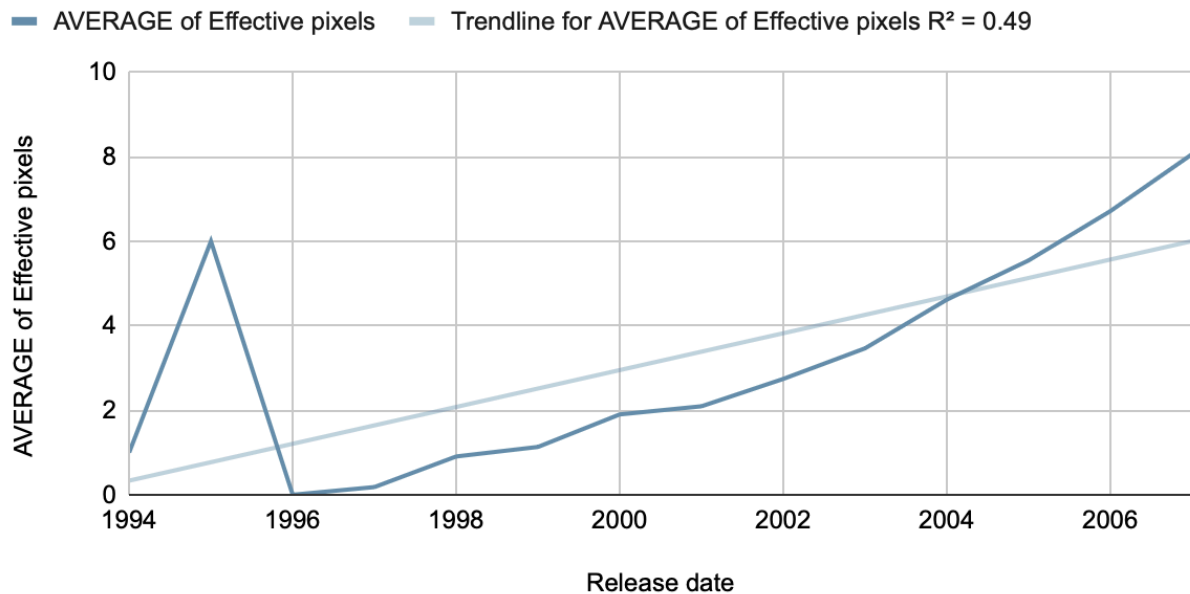


Figure 2

Findings

The average effective megapixel count increased substantially between 1994 and 2007, indicating a clear upward trend in camera resolution over time. While the averages for 1994 through 1996 fluctuate due to the limited number of cameras available in the dataset during those years, the trend becomes much more consistent beginning in 1998. From that point forward, average camera resolution steadily increased, demonstrating significant technological advancements in digital camera image quality throughout the study period.

Business Question 3

Do heavier cameras tend to have higher effective megapixel counts?

Camera Weight vs. Effective Megapixels

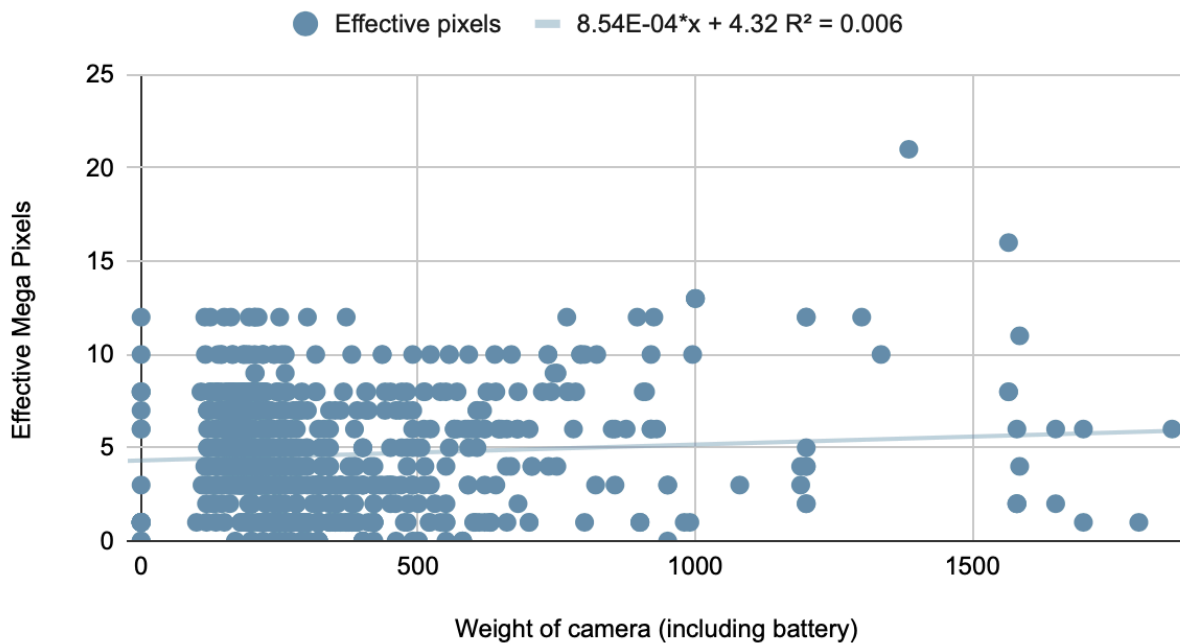


Figure 3

Findings

The scatter plot suggests a very weak positive relationship between camera weight and effective megapixel count. Although heavier cameras occasionally have higher resolutions, the points are widely dispersed, indicating that camera weight alone is not a strong predictor of effective megapixels. This suggests that other design and technological factors influenced camera resolution more than physical weight during the study period.

Business Question 4

How did average camera weight change between 1994 and 2007?

AVERAGE Weight (inc. batteries) by Release date

(1994-2007)

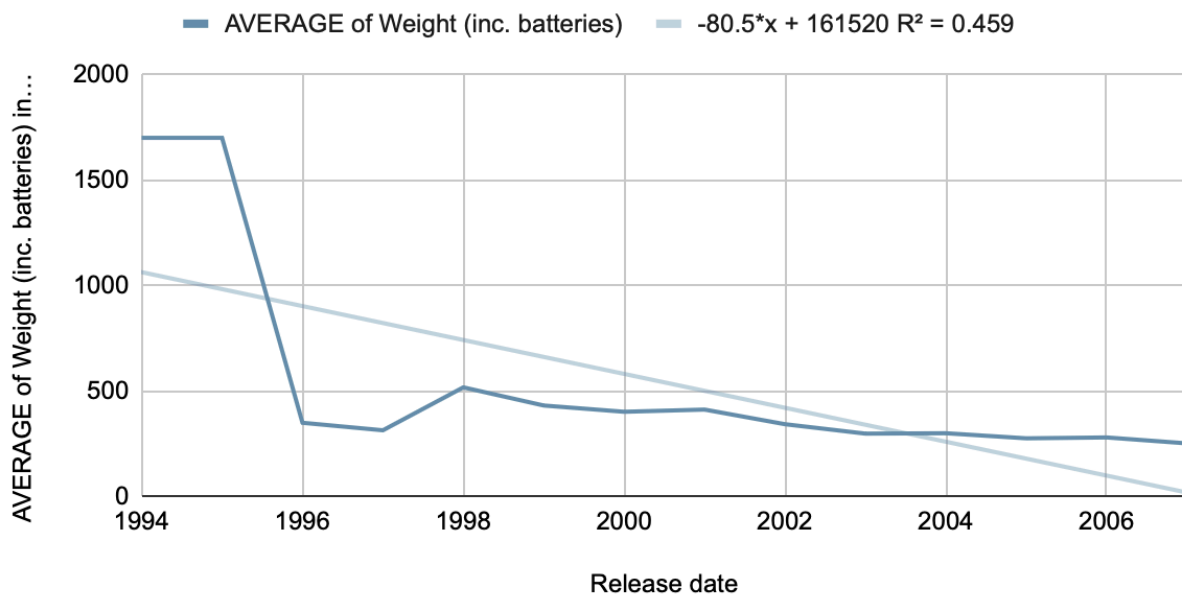


Figure 4

Findings

The average camera weight shows a downward trend between 1994 and 2007. Cameras in the dataset were heaviest during the earliest years, averaging approximately 1,700 grams in 1994 and 1995. Average weight declined sharply after 1995, experienced a slight increase around 1998, and then continued a gradual downward trend through 2007. Overall, the results suggest that digital cameras became progressively lighter over time as technology advanced.

Business Question 5

Which manufacturers produced the heaviest cameras on average between 1994 and 2007?

AVERAGE Camera Weight by Manufacturer

(1994-2007)

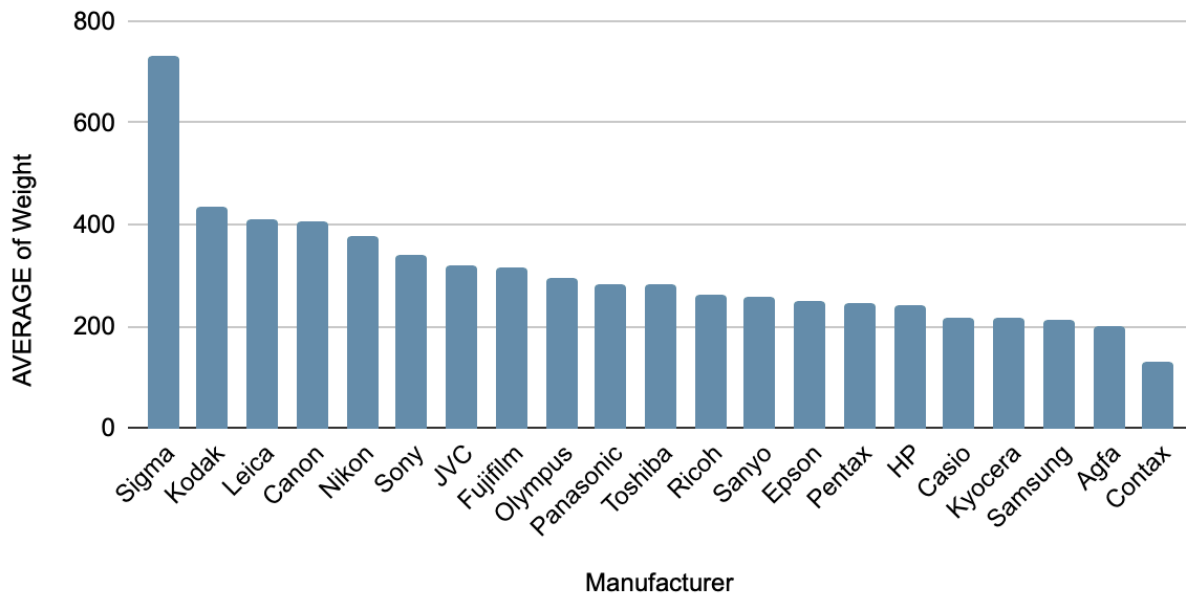


Figure 5

Findings

Sigma produced the heaviest cameras on average, with an average weight of approximately 730 grams, followed by Kodak at approximately 436 grams. Average camera weight then gradually declined across the remaining manufacturers. Contax produced the lightest cameras on average, with an average weight of approximately 130 grams. Most manufacturers averaged between 200 and 400 grams, while Sigma stood out as a notable outlier with substantially heavier camera models.

Key Findings

- Camera resolution improved substantially between 1994 and 2007, with average effective megapixels showing a clear upward trend over the study period.
- Average camera weight generally decreased over time, suggesting that manufacturers produced increasingly compact cameras while continuing to improve imaging technology.
- Samsung had the highest average effective megapixel count among the manufacturers included in the dataset, while Agfa had the lowest.
- Sigma produced the heaviest cameras on average, whereas Contax produced the lightest cameras on average.
- Camera weight showed only a very weak positive relationship with effective megapixels, indicating that heavier cameras did not necessarily have higher image resolution.

Limitations & Conclusion

This analysis has several limitations that should be considered when interpreting the findings:

- The number of cameras included in the dataset varies significantly by release year. For example, 1994 and 1995 each contain only one camera, while later years contain more than 100 cameras. As a result, averages for the earliest years are more susceptible to being influenced by individual camera models.
- The dataset did not include a clearly defined and reliable price variable. During exploratory analysis, repeated price values and comparisons with historical camera specifications suggested that the variable did not represent original retail prices. To maintain the integrity of the analysis, the price variable was excluded from the final business questions and conclusions.
- The analysis was limited to the information available in the dataset. Other factors that may influence camera design and technological advancement, such as sensor size, market segment, or production volume, were not examined.

This analysis examined how digital camera specifications evolved between 1994 and 2007 and how manufacturers differed during this period. The results indicate that digital camera technology advanced considerably, with average effective megapixel counts increasing steadily while average camera weight generally decreased over time. Comparisons among manufacturers revealed differences in both average resolution and camera weight, although many manufacturers produced cameras with similar overall specifications. Additionally, the weak relationship between camera weight and effective megapixels suggests that larger cameras did not necessarily

offer higher image resolution. Despite limitations within the dataset, including an unreliable price variable and smaller sample sizes in the earliest years, the analysis successfully identified meaningful trends in digital camera development and demonstrated the application of data cleaning, exploratory data analysis, data visualization, and interpretation to answer business questions.

References

Crawford. (n.d.). *1000 Cameras Dataset*. Kaggle. Retrieved July 8, 2026, <https://www.kaggle.com/datasets/crawford/1000-cameras-dataset>